

Gradient of n colors ranging from color 1 and color 2

I often work with `ggplot2` that makes gradients nice ([click here for an example](#)). I have a need to work in base and I think `scales` can be used there to create color gradients as well but I'm severely off the mark on how. The basic goal is generate a palette of n colors that ranges from x color to y color. The solution needs to work in base though. This was a starting point but there's no place to input an n.

```
scale_colour_gradientn(colours=c("red", "blue"))
```

I am well aware of:

```
brewer.pal(8, "Spectral")
```

from `RColorBrewer`. I'm looking more for the approach similar to how `ggplot2` handles gradients that says I have these two colors and I want 15 colors along the way. How can I do that?

r

asked Nov 12 '12 at 23:16



[Tyler Rinker](#)

51.5k 25 169 331

1 I think you need the **scales** package; the latest **ggplot2** versions depend on this for the underlying code. I.e. you don't need **ggplot2** to use **scales**, you just need the **scales** package. No idea how the functions in **scales** works though :-). – [Gavin Simpson](#) Nov 12 '12 at 23:25

4 Answers

`colorRampPalette` could be your friend here:

```
colfunc <- colorRampPalette(c("black", "white"))
colfunc(10)
# [1] "#000000" "#1C1C1C" "#383838" "#555555" "#717171" "#8D8D8D" "#AAAAAA"
# [8] "#C6C6C6" "#E2E2E2" "#FFFFFF"
```

And just to show it works:

```
plot(rep(1,10),col=colfunc(10),pch=19,cex=3)
```



edited Apr 30 '14 at 23:05

answered Nov 12 '12 at 23:21



[thelatemail](#)

53.1k 6 60 120

+1 beaten by 8 seconds! – [mnel](#) Nov 12 '12 at 23:22

1 @mnel - quickdraw win! ;-) – [thelatemail](#) Nov 12 '12 at 23:23

1 That is beautiful. – [Tyler Rinker](#) Nov 12 '12 at 23:34

6 Note that if you're particularly enamoured with a pre-existing palette, e.g. `brewer.pal(8, "Spectral")`, you can give the resulting vector of colours to `colorRampPalette` to generate more colours along that ramp. For example: `colorRampPalette(brewer.pal(8, "Spectral"))`. – [jbaums](#) Apr 30 '14 at 23:15

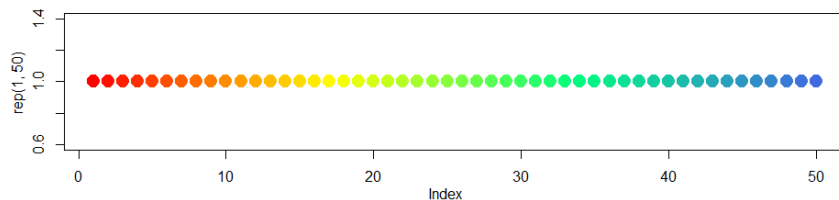
What about the color with the diagonoal lines or shapes in the block ? @thelatemail – [user3631848](#) May 16 '16 at 8:46

Just to expand on the previous answer `colorRampPalette` can handle more than two colors.

So for a more expanded "heat map" type look you can....

```
colfunc<-colorRampPalette(c("red","yellow","springgreen","royalblue"))
plot(rep(1,50),col=(colfunc(50)), pch=19,cex=2)
```

The resulting image:



edited Aug 26 '15 at 14:13

Veerendra Gadekar
3,846 10 20

answered May 29 '15 at 15:41

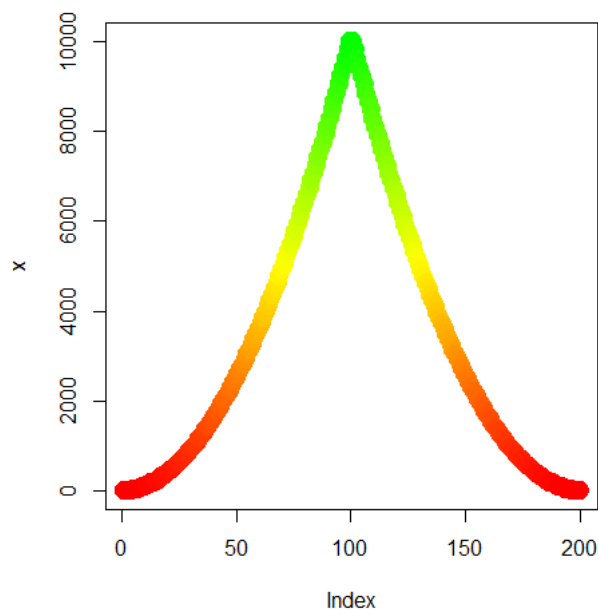
jsol
233 2 5

can I also specify the diagonal lines or shapes in the color ? @jsol – user3631848 May 16 '16 at 8:47

I like this palette but there is way too much green and very little yellow. Is there a way to correct this? – user4050 May 19 at 10:02

Try the following:

```
color.gradient <- function(x, colors=c("red","yellow","green"), colsteps=100) {
  return( colorRampPalette(colors) (colsteps) [ findInterval(x, seq(min(x),max(x),
length.out=colsteps)) ] )
}
x <- c((1:100)^2, (100:1)^2)
plot(x,col=color.gradient(x), pch=19,cex=2)
```



answered Feb 26 '16 at 16:04

Daniel Hoop
78 2 7

The above answer is useful but in graphs, it is difficult to distinguish between darker gradients of black. One alternative I found is to use gradients of gray colors as follows

```
palette(gray.colors(10, 0.9, 0.4))
plot(rep(1,10),col=1:10,pch=19,cex=3))
```

More info on gray scale [here](#).

Added

When I used the code above for different colours like blue and black, the gradients were not that clear. `heat.colors()` seems more useful.

[This document](#) has more detailed information and options. pdf

edited Jan 28 '15 at 3:08

thelatemail
53.1k 6 60 120

answered Sep 24 '14 at 11:30

Anusha
791 1 14 19

1 I think this answer is superior for black to white but is not generalizable to colors. Thank you for adding this valuable information. +1 – [Tyler Rinker](#) Sep 24 '14 at 12:52

Added a link which provides better options for color gradients and hues which work in both color and B&W. – [Anusha](#) Sep 24 '14 at 15:40

Nice Link! Very informative. – [Tyler Rinker](#) Sep 24 '14 at 16:00

Is there an update to this link.. As of 1/2015, it's broken. – [David DelMonte](#) Jan 13 '15 at 12:41

2 @DavidDelMonte - [web.archive.org/web/20141111182737/http://www.stat.tamu.edu/...](http://web.archive.org/web/20141111182737/http://www.stat.tamu.edu/) check archive.org first always. – [thelatemail](#) Jan 28 '15 at 3:07

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